NEWTON SCHOOL

PRE-CALCULUS | HOMEWORK

Inverse Trig Functions: Domain, Range & Evaluation

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Standard: *CCSS.MATH.CONTENT.HSF-BF.B.4 — Find inverse functions. Solve equations of the form $f(x) = c$ for a simple function f that has an inverse and write an expression for the inverse. Evaluate and interpret inverse trigonometric expressions using domain and range constraints.*

🔔 **Mr. Kenny's Key Point**

Domain & Range determine whether you can just "cancel" inverse and trig — or whether you need to **reduce the angle** into the valid range first.

$\sin^{-1}$: domain $[-1, 1]$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$ · $\cos^{-1}$: domain $[-1, 1]$, range $[0, \pi]$ · $\tan^{-1}$: domain $(-\infty, \infty)$, range $(-\frac{\pi}{2}, \frac{\pi}{2})$

Reference Notes — Inverse Trig: Domain, Range & Key Rules

Function	Domain (input)	Range (output angle)	Special values
$\sin^{-1}(x)$	$[-1,\,1]$	$\left[-\dfrac{\pi}{2},\,\dfrac{\pi}{2}\right]$	$\sin^{-1}(0)=0,\ \sin^{-1}(1)=\frac{\pi}{2},\ \sin^{-1}(-1)=-\frac{\pi}{2}$
$\cos^{-1}(x)$	$[-1,\,1]$	$[0,\,\pi]$	$\cos^{-1}(0)=\frac{\pi}{2},\ \cos^{-1}(1)=0,\ \cos^{-1}(-1)=\pi$
$\tan^{-1}(x)$	$(-\infty,\,\infty)$	$\left(-\dfrac{\pi}{2},\,\dfrac{\pi}{2}\right)$	$\tan^{-1}(0)=0,\ \tan^{-1}(1)=\frac{\pi}{4},\ \tan^{-1}(-1)=-\frac{\pi}{4}$

Reference Triangle Method: Draw a right triangle from the *inside* arc-function, then evaluate the *outside* trig. E.g. for $\cos(\sin^{-1}(\frac{3}{5}))$: let $\theta=\sin^{-1}(\frac{3}{5})$, so $\sin\theta=\frac{3}{5}$. Triangle: opp = 3, hyp = 5, adj = 4. Then $\cos\theta=\frac{4}{5}$.

Reduction Rule (range-aware): $\sin^{-1}(\sin\theta)=\theta$ *only if* $\theta\in[-\frac{\pi}{2},\frac{\pi}{2}]$. If θ is *outside* the range, find the angle *inside* the range with the same sine value. Example: $\sin^{-1}(\sin(\frac{2\pi}{3}))=\sin^{-1}(\frac{\sqrt{3}}{2})=\frac{\pi}{3}$ (not $\frac{2\pi}{3}$).

Identity: $\sin^{-1}(x)+\cos^{-1}(x)=\dfrac{\pi}{2}$ for all $x\in[-1,1]$.

Tip: Always check: is the angle in the *output range* of the inverse function? If no → reduce first.

Part A — Composite Evaluation Using Reference Triangles

Use the reference triangle method. Draw a right triangle based on the inner inverse trig expression, label all three sides, then evaluate the outer trig function. **Show all work.**

1. Find the exact value of $\cos\left(\sin^{-1}\left(\dfrac{3}{5}\right)\right)$.

2. Find the exact value of $\sin\left(\cos^{-1}\left(\frac{2}{3}\right)\right)$.

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3. Find the exact value of $\tan\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$.

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4. Find the exact value of $\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right)$.

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5. Write a simplified expression for $\sin\left(\tan^{-1}\left(\frac{x}{2}\right)\right)$ in terms of x.

6. Find the exact value of $\sec\left(\cos^{-1}\left(\frac{1}{4}\right)\right)$. *(Hint: recall* $\sec\theta = \dfrac{1}{\cos\theta}$*.)*

Part B — Range-Aware Reduction

Evaluate each expression carefully. Check whether the angle is inside the valid output range before simplifying. If it is outside the range, reduce it to an equivalent angle inside the range. **Show your reasoning.**

7. Evaluate $\sin^{-1}\left(\sin\left(\frac{\pi}{3}\right)\right)$.

8. Evaluate $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$. *(Is* $\frac{2\pi}{3}$ *in the range of* $\sin^{-1}$*?)*

9. Evaluate $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$.

10. Evaluate $\tan^{-1}(\tan(\pi))$. *(Note: π is outside the range of $\tan^{-1}$.)*

11. Evaluate $\sin^{-1}\left(\cos\left(\frac{3\pi}{2}\right)\right)$.

Part C — Double Angle & Advanced Identities

Apply double-angle or sum formulas. Let θ equal the inner inverse trig expression, build a reference triangle, then use the relevant identity. **Show every step.**

12. Find the exact value of $\tan\left(2\sin^{-1}\left(\frac{1}{3}\right)\right)$. *(Use the double-angle formula:* $\tan(2\theta) = \dfrac{2\tan\theta}{1-\tan^2\theta}$*.)*

13. Find the exact value of $\cos\left(2\tan^{-1}\left(\frac{1}{2}\right)\right)$. *(Useful form:* $\cos(2\theta) = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$*.)*

14. Find the exact value of $\sin\big(\sin^{-1}(0.8) + \cos^{-1}(0.6)\big)$. *(Let* $\alpha = \sin^{-1}(0.8)$ *and* $\beta = \cos^{-1}(0.6)$. *Use* $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$.*)*

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15. Prove that $\sin^{-1}(x) + \cos^{-1}(x) = \dfrac{\pi}{2}$ for all $x \in [-1, 1]$. *(Hint: Let* $\theta = \sin^{-1}(x)$ *and show* $\cos^{-1}(x) = \frac{\pi}{2} - \theta$.*)*
